

Program:

```
#include<stdio.h>
// Function to swap two elements in the array
voidswap(int* a, int* b) {
int temp = *a;
*a = *b;
*b = temp;
}
// Function to partition the array and return the index of the pivot
intpartition(int arr[], int low, int high) {
int pivot = arr[high];
// Choosing the last element as the pivot
int i = low - 1;
// Index of smaller element
for (int j = low; j < high; j++) {
// If current element is smaller than the pivot
if (arr[j] < pivot) {
i++;
swap(&arr[i], &arr[j]);
}
}
// Place the pivot element in its correct position
swap(&arr[i + 1], &arr[high]);
return i + 1;
}
// Function to implement Quick Sort
voidquickSort(int arr[], int low, int high) {
if (low < high) {
// Partitioning index
int pi = partition(arr, low, high);
// Recursive sorting of the sub-arrays
quickSort(arr, low, pi - 1);
quickSort(arr, pi + 1, high);
}
}
// Function to print the array
voidprintArray(int arr[], int size) {
for (int i = 0; i < size; i++)
printf("%d ", arr[i]);
printf("\n");
}
// Driver program
intmain() {
```

```
int arr[] = {12, 7, 11, 13, 5, 6};  
int n = sizeof(arr) / sizeof(arr[0]);  
printf("Original array:\n");  
printArray(arr, n);  
quickSort(arr, 0, n - 1);  
printf("Sorted array:\n");  
printArray(arr, n);  
return 0;  
}
```